Unit 5: Thermodynamics, Radiation, Oscillations and Cosmology

5.3 Thermodynamics

This topic covers specific heat capacity, specific latent heat, internal energy and the gas equation.

This topic may be studied using applications that relate, for example, to space technology.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

c is the specific heat capacity
L is the specific latent heat

125	be able to use the equations $\Delta E = mc\Delta\theta$ and $\Delta E = L\Delta m$
	The ***specific heat capacity*** **(c)** is: ***"The amount of energy required to increase the temperature of 1 kg of a substance by 1°C (or) by 1K, without changing its state"*** You can measure the amount of energy required to change the temperature of a substance using the following formula: $\boldsymbol{\Delta E = mc\Delta\theta}$ The ***specific latent heat (L)*** is: ***"The amount of energy required to change the state of 1 kg of material, without changing its temperature"*** You can measure the amount of energy required to change the state of a substance using the following formula: $\boldsymbol{\Delta E = L\Delta m}$ There are two types of specific latent heat: 1. Specific latent heat of *fusion* (when solid changes to liquid) (melting) 2. Specific latent heat of *vaporization* (when liquid changes to gas) (boiling)
126	**CORE PRACTICAL 12: Calibrate a thermistor in a potential divider circuit as a thermostat**
127	**CORE PRACTICAL 13: Determine the specific latent heat of a phase change**
128	understand the concept of *internal energy* as the random distribution of potential and kinetic energy amongst molecules
	The **internal energy (U)** of a substance is defined as: ***"The sum of the randomly distributed kinetic and potential energies of all its particles within a given mass of a substance"*** Particles are *randomly distributed*, meaning they have different *speeds* and *separations* (meaning the *kinetic* and *potential* energies of a body are randomly distributed) The *internal energy* of a system is determined by: 1. **Temperature** (higher temperature, higher kinetic energy and vice versa) 2. **Random motion** of molecules 3. **Phase of matter** (gases have the highest internal energy, solids have the lowest) 4. **Intermolecular forces** between the particles (greater intermolecular forces, higher potential energy and vice versa) - this is linked to the phase (solid, liquid, gas) that the matter is in The internal energy of a system can increase by: - **Doing work** on it - **Adding heat** to it The internal energy of a system can decrease by: - **Losing heat** to its surroundings - **Changing state** from a gas to a liquid or a liquid to a solid (because potential energy decreases while kinetic energy is kept constant)

KE is due to speed of molecules and gives the material its temperature
PE is due to separation between molecules and their position within the structure

129	understand the concept of *absolute zero* and how the average kinetic energy of molecules is related to the absolute temperature

Absolute zero (−273°C or 0 K) is the lowest possible temperature, and is
"The temperature at which *particles have no kinetic energy and the volume and pressure of a gas are zero*"

- The absolute scale of temperature is the kelvin scale
- A change of 1 K is equal to a change of 1°C
- To convert between the two you can use the formula: $K = C + 273$

The average kinetic energy of molecules and their absolute temperature are related by the following equation:

$$\text{Average kinetic energy of one molecule of gas} = \tfrac{1}{2} m\langle c^2\rangle = 3/2\ kT$$

Where T is temperature in Kelvin and k is the Boltzmann constant

From this equation, we can see that the average kinetic energy of molecules is directly proportional to temperature (in Kelvin)

130	be able to use the equation $pV = NkT$ for an ideal gas

Boltzmann constant *k* is defined as:

$$k = \frac{R}{N_A}$$

Where R = molar gas constant and N_A = Avogadro's constant

In an ideal gas, there is no other interaction other than perfectly elastic collisions between the gas molecules, which shows that no intermolecular forces act between molecules

As potential energy is associated with intermolecular forces, an ideal gas has no potential energy, therefore:

"An ideal gas's internal energy is equal to sum of the kinetic energies of all of its particles"

The Gas Laws

1. **Boyle's Law**

(pressure is inversely proportional to the volume of a gas, where temperature is constant)

$$p_1V_1 = p_2V_2$$

In Boyle's law, since temperature is constant, the mean KE of the molecules is also constant

2. **Charles's Law**

(volume is proportional to the temperature of a gas, where pressure is constant)

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

3. **Pressure Law**

(pressure is proportional to the temperature, where volume is constant)

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

These three laws combine to form the ideal gas equation: $\boxed{pV = NkT}$

Where pressure (*p*), volume of ideal gas (*V*), number of molecules (*N*) and absolute temperature (*T*)

131	**CORE PRACTICAL 14: Investigate the relationship between pressure and volume of a gas at fixed temperature**

132	be able to derive and use the equation $\frac{1}{2} m\langle c^2\rangle = \frac{3}{2} kT$
	The kinetic theory model equation relates several features of a fixed mass of gas, including its pressure, volume and mean kinetic energy with assumptions made: • No intermolecular forces act on the molecules • The duration of collisions is negligible in comparison to time between collisions • The motion of molecules is random, and they experience perfectly elastic collisions • The motion of the molecules follows Newton's laws • The molecules move in straight lines between collisions

Derivation: $\frac{1}{2} m\langle c^2\rangle = \frac{3}{2} kT$

Where c is avg speed of a molecule

By using the kinetic theory model equation and the ideal gas equation (from above), you can **derive** the equation for the average kinetic energy of a molecule (mentioned in 5.3.148).

$$pV = \frac{1}{3}Nm\langle c^2\rangle \qquad\qquad pV = NkT$$

First, set the above equations equal to each other and simplify.

$$\frac{1}{3}Nm\langle c^2\rangle = NkT$$

$$\frac{1}{3}m\langle c^2\rangle = kT$$

Finally, make the left hand side of the equation resemble the equation for the kinetic energy of an object ($E_k = \frac{1}{2} mv^2$), by multiplying both sides by 3/2.

$$\frac{3}{2} \times \frac{1}{3}m\langle c^2\rangle = \frac{3}{2}kT$$

$$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$$

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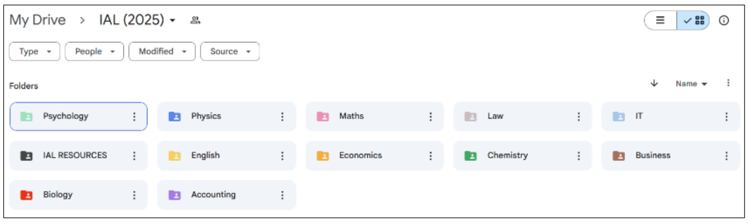

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